References:
1. Introduction to Industrial Organization by L. M. B. Cabral
2. The Economics of Industrial Organization by W. G. Shepherd
3. The Theory of Industrial Organization by Jean Tirol

Chapter-5: Monopoly and Price Discrimination

Chapter Outline:

-Definition, Conditions

-Merger, Motives

❑Q. Price Discrimination.

Answer: The monopolist has control over pricing, demand, and supply decisions, thus, sets prices in a way, so that maximum profit can be earned. The monopolist often charges different prices from different consumers for the same product. This practice of charging different prices for identical product is called price discrimination.

According to Robinson, "Price discrimination is charging different prices for the same product or same price for the differentiated product."

❑Q. Conditions of the Price discrimination

Answer:

- **Existence of Monopoly:** A supplier can discriminate prices only when there is monopoly. The degree of the price discrimination depends upon the degree of monopoly in the market.

- **Separate Market:** There must be two or more markets that can be easily separated for discriminating prices. The buyer of one market cannot move to another market and goods sold in one market cannot be resold in another market.

- **No Contact between Buyers:** A supplier can discriminate prices if there is no contact between buyers of different markets. If buyers in one market come to know that prices charged in another market are lower, they will prefer to buy it in other market and sell in own market. The monopolists should be able to separate markets and avoid reselling in these markets.

- **Different Elasticity of Demand:** The elasticity of demand in the markets should differ from each other. In markets with high elasticity of demand, low price will be charged, whereas in markets with low elasticity of demand, high prices will be charged. Price discrimination fails in case of markets having same elasticity- of demand.

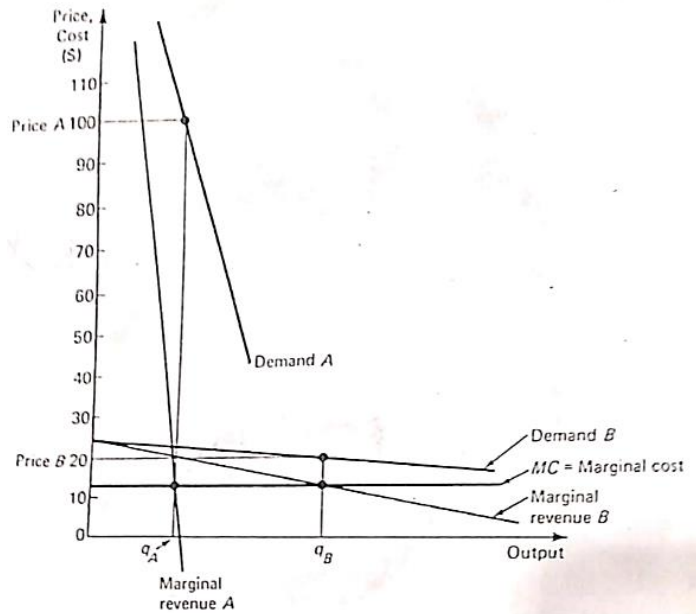
❑Q. The Analysis of Price discrimination

Answer: There are two cases:

a. **Identical Costs, Differing Prices**;
b. **Price Differing Less Than Costs**

Mohammed Mohiuddin,
Assistant Professor, Department of Economics, University of Chittagong, Chattogram
Adjunct Faculty, 79 BMA L/C; ECON:2405
mail: mohiuddin_eco@cu.ac.bd

a. Identical Costs, Differing Prices: Consider a constant and identical MC curve for two customers groups.



Group-A: Inelastic Demand and Group-B: Highly Elastic Demand
Price for Group-A= \$100 and Price for Group-B=\$20
Hence, the price-cost ratio becomes

$$\text{Price } A \,/\, \text{Marginal cost}_A > \text{Price } B \,/\, \text{Marginal cost}_B$$

Figure 12.1 can also illustrate the pricing of a lifesaving drug. The same drug is sold to two groups: (1) to druggists for resale to individuals through doctors' prescriptions; and (2) to large hospitals, for dispensing to patients. The druggists' customers have *low* demand elasticity, for they merely buy what their doctor writes on their prescriptions. By contrast, the hospitals have *elastic* demand. They can bargain shrewdly, playing off the drug companies against one another to get a low price. Thus the identical drug, costing perhaps \$3 per dozen pills to make, might sell for \$16 per dozen to retail druggists and \$5 per dozen t...

Price Differing Less Than Costs: Consider

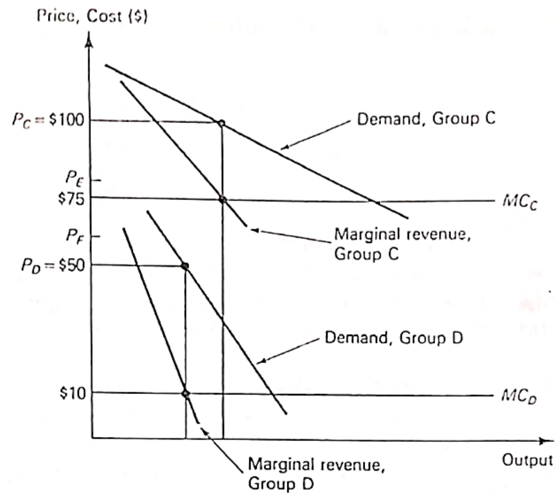
MC_C= Guaranteed Airline Seat and Services=\$75

MC_D= Standby Airline Seat and Services if empty seat=\$10

Mohammed Mohiuddin,
Assistant Professor, Department of Economics, University of Chittagong, Chattogram
Adjunct Faculty, 79 BMA L/C; ECON:2405
mail: mohiuddin_eco@cu.ac.bd

one more body.

In deciding on its prices, the airline seeks to maximize its profits from each of the two groups; it sets output for each where marginal revenue equals marginal cost. That results in a fare of $100 for regular passengers and $50 for standby passengers, as shown in Figure 12.2. A regular passenger might be irritated to see a standby rider pay only half fare, but in fact *the discrimination runs the other way*. It is the standby passengers in group D who are being discriminated against, because the ratio of price to cost for that group is higher than group C's ratio. The price-cost ratios are exactly 4 to 3 for group C and 5 to 1 for group D.



$\left(\frac{P_C}{MC_C} = 100/75, \frac{P_D}{MC_D} = 5/1\right)$

If the firm instead chose a single price for both groups in Figure 12.2, that, too, would be discrimination. Price P_E is such a single discriminatory price. (The price-cost ratios would be about 1.1 for the buyers in group C and 5.0 for those in group D.) Price P_F is another. (The ratios are now 0.9 and 4.0.) P_F is especially discriminatory because group C buyers now pay even less than the cost of supplying them. Both P_E and P_F are irrational as shown, for they do not maximize the firm's profits. Marginal revenue is not in line with marginal cost for both groups. But there may be occasions when a firm will indeed set price below cost; such "predatory" pricing will be discussed shortly.

Discrimination is intensive when (1) demand elasticities vary sharply among customers, and (2) the barriers to reselling by customers are high. Elasticities vary

Mohammed Mohiuddin,
Assistant Professor, Department of Economics, University of Chittagong, Chattogram
Adjunct Faculty, 79 BMA L/C; ECON:2405
mail: mohiuddin_eco@cu.ac.bd